

A Hilbert-based Coherence Factor for Photoacoustic Imaging

Soheil Hakakzadeh

Department of Electrical Engineering
Sharif University of Technology

Tehran, Iran

soheilh@ieee.org

Research and Development Department, Med Fanavaran Plus
Co., Karaj, Iran.

Zahra Kavehvash

Department of Electrical Engineering
Sharif University of Technology

Tehran, Iran

kavehvash@sharif.edu

Research and Development Department, Med Fanavaran Plus
Co., Karaj, Iran

Abstract— Among the different configurations of a photoacoustic imaging (PAI) system, the planar configuration has many applications. Due to the limited-view of planar configuration, the resolution and contrast of the imaging system are low. To improve resolution and contrast, the coherence factor (CF) can be applied to the reconstructed image. Although CF improves the resolution and contrast, it causes missed data in the reconstructed image. In this paper, we proposed a Hilbert-based CF (H-CF) method to compensate for the missed data issue and also increase the contrast. Three numerical studies were conducted to evaluate the proposed method's performance. The numerical results show that applying H-CF increases contrast ratio (CR), and structural similarity index measure (SSIM) up to 0.19 and 19 dB, respectively. Therefore, the proposed method can be a suitable replacement for conventional methods.

Keywords—Hilbert transform, Coherence factor, Beamforming, Photoacoustic, Image reconstruction.

I. INTRODUCTION

Recently, a new imaging modality based on the photoacoustic effect has been developed for biomedical imaging. The photoacoustic effect was discovered by Graham Bell in 1880 [1]. Then in the first decade of the 21st century, this effect evolved and the new imaging modality called photoacoustic imaging (PAI) was developed by Wang in biomedicine [2]. In a PAI system, first, a laser source irradiates the biological tissue. Then the biological tissue, based on its optical absorption coefficient, absorbs the energy of the irradiated light. After that, based on the absorption of the light energy and thermal expansion phenomena, acoustic waves are generated. Finally, the reconstructed image will be obtained by performing a mathematical algorithm on the captured acoustic signal by ultrasound transducers. The procedure of a PAI system is shown in Fig. 1.

Among many conventional photoacoustic reconstruction algorithms, the back-projection algorithm based on delay-and-sum (DAS) is widely used in PAI systems [3, 4]. The transducer array elements in a PAI system are non-directional. Therefore, uni-directional back-projection should be considered in the reconstruction algorithm. The uni-directional back-projection method results in appearing strike artifacts in the reconstructed image [5]. In the planar transducer configuration of PAI systems, transducers could not capture all solid angles of each generated acoustic wave [6, 7]. Therefore in comparison to circular transducer configuration, the values of mentioned strike artifacts are more [8]. Moreover, capturing limited solid angles causes degradation of the lateral resolution in the reconstructed image. Still, due to some practical limitations, in many applications, using the circular configuration is impractical

[9]. Hence, linear-based configurations play a critical role in PAI imaging, especially in hand-held PAI systems.

To suppress strike artifacts and improve the image quality in a PAI system, some weighting factors and reconstruction algorithms have been presented. An Adapted delay-and-sum (ADAS) algorithm is developed to enhance the contrast of the reconstructed image [10]. Delay multiply and sum (DMAS) and filtered DMAS algorithms are also presented to improve tangential resolution and suppress artifacts [11]. Since the computational cost of DMAS is high, which may be impractical in some applications, Jeon, et al. [12] proposed the real-time version of the DMAS algorithm. Adaptive beamformers such as the minimum variance (MV) method are presented to improve resolution [13]. However, the improvement of the MV algorithm in a crowded and high detailed media may not be significant [14]. To suppress artifacts in circular and planar configurations, standard deviation-based factor and unipolar back-projection algorithm are presented, respectively [15, 16]. Some model-based and deep learning-based reconstruction methods are also proposed to enhance resolution and suppress artifacts [17, 18]. However, these methods may fail in heterogeneous and crowded media, especially in *in-vivo*. A PA signal restoration algorithm is presented to improve axial resolution and artifacts by compensating the transducer band-limited effect [19]. But even by using this method, some strike artifacts remain. Coherence weighting factor (CF) have been proposed to suppress artifacts and improve resolution [4, 14, 20]. Based on the simplicity of CF, the computational complexity would be low. Since the captured PA wave by the transducers is bipolar and CF is unipolar, applying CF to a conventional DAS algorithm may cause missing data.

In this paper, to suppress the strike artifacts and improve resolution in a planar configuration PAI system, a modified version of CF here called Hilbert-based CF (H-CF) is applied at the end of the DAS algorithm. Unlike the conventional CF method, the results of the H-CF show there are no missing data appearing in the reconstructed image. Also, a high-quality reconstructed image in terms of resolution, signal-to-noise ratio (SNR), contrast ratio (CR), and structural similarity index measure (SSIM) can be obtained by applying H-CF at the end of the DAS algorithm.

In section II, the theory of DAS and CF are presented. Section III contains the proposed method. Section IV is the simulation study, results, and discussion. Also, section V is the conclusion.

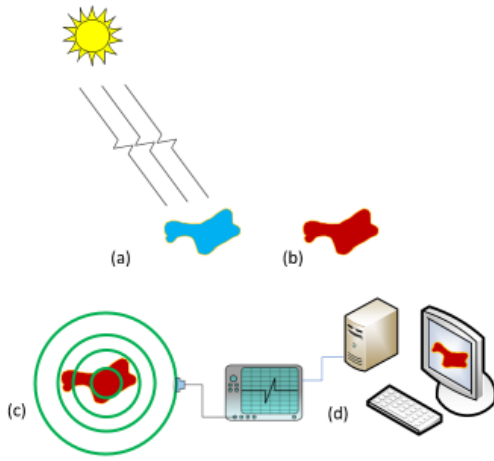


Fig. 1: The schematic of the photoacoustic procedure in a PAI system. (a) laser source irradiates the PA absorber. (b) thermal expansion due to light irradiation. (c) ultrasound wave generation and the transducer captures the ultrasound wave (PA signal). (d) post-processing and image reconstruction process.

II. BACKGROUND

As the PA absorber is heated by the laser source, based on the optical absorption coefficient, an ultrasound wave will be generated in all directions. Therefore based on the time of flight (TOF) between the PA absorber and elements of the transducer array, all transducers capture the generated wave in a specific TOF. To reconstruct the map of PA absorbers, the delay between each pixel (x, y) in the region of interest (ROI) and all of the transducer elements should be calculated $(\Delta t_{(x,y)})$. Then, by summing the related value based on the TOF of captured PA signal of all transducers, the initial value of generated PA signal, which represents the absorption coefficient of that PA absorber, will be obtained. The mathematical relation for the DAS algorithm on pixel (x, y) is as follows [4]:

$$p_{DAS}^{(b)}(x, y) = \sum_{i=1}^N p_n(\Delta t_{(x,y)}) \quad (1)$$

where, N is the number of transducer elements, $p_n(\Delta t_{(x,y)})$ is related pressure value based on the TOF of captured PA signal of n^{th} transducer. $p_{DAS}^{(b)}(x, y)$ is the output of the DAS algorithm for pixel (x, y) .

The coherence factor (CF) for the pixel in (x, y) in ROI, is the sum of the squares of related pressure values based on the TOF of all transducers $(p_n(\Delta t_{(x,y)}))$ over the square of sum of all related pressure values which is the coherency to the incoherency ratio of each pixel in the ROI which is given in (2):

$$CF(x, y) = \frac{\sum_{n=1}^N |p_n(\Delta t_{(x,y)})|^2}{N |\sum_{n=1}^N p_n(\Delta t_{(x,y)})|^2} \quad (2)$$

Pixel-by-pixel multiplication of expression (1) and (2) yields the final coherence-based reconstructed image (DAS-CF) which is as follows:

$$p_{DAS-CF}(x, y) = p_{DAS}^{(b)}(x, y) \times CF(x, y) \quad (3)$$

III. PROPOSED METHOD

The optical absorption coefficients of biological tissue are non-negative. However, the propagation of the generated acoustic wave would be bipolar. It means that the acoustic wave contains both positive and negative amplitudes (see the example of the captured wave by the transducer in Fig 1. (c)). This bipolar acoustic wave results in a bipolar reconstructed image. To solve the bipolar issue, two stages of post-processing are appropriate; 1) applying Hilbert transformation on the output of the reconstructed image. 2) Since the output of the Hilbert transform contains complex values, computing the absolute values should be needed. Applying Hilbert transform extracts the envelope of the bipolar reconstructed signal (reconstructed image) and results in a unipolar reconstructed image.

In the CF equation, both the numerator and denominator of the fraction are positive. Therefore, CF results in a unipolar factor. Since acoustic signals contain all positive, zero, and negative components (see Fig. 1(c)) and the CF equation calculates absolute values, the CF results would contain positive and zeros values. Therefore, some pixels may be positive in the output of the DAS algorithm and the same pixels may be zero in the output of the CF factor. These zero pixels will remain in the product of the DAS algorithm and CF factor. This zero-value issue, results in missing data in the final CF-based reconstructed image.

Here, to tackle the zero value issue of CF result, instead of applying Hilbert transform to the reconstructed image, Hilbert transform is applied to the captured acoustic signals by transducers $(p_n(.))$ and the result $(p_{Hil,n}(.))$ will be used as the input of both DAS algorithm and CF factor and is given in (4).

$$p_{Hil,n}(t) = HT(p_n(t)), \quad (4)$$

where, $HT(.)$ is the Hilbert transform operator which is given as follows:

$$HT(X(t)) = \hat{X}(t) = \frac{1}{\pi t} * X(t), \quad (5)$$

where, $*$ is the convolution operator. In (5) assumed that the $X(t)$ is real and has no DC component. By substituting (4) in (1), new DAS result of (x, y) pixel will be obtained, which is given in (6).

$$p_{Hil-DAS}^{(b)}(x, y) = |\sum_{i=1}^N p_{Hil,n}(\Delta t_{(x,y)})|, \quad (6)$$

where, $|\cdot|$ is the absolute value operator.

By substituting (4) in (2), new CF result of (x, y) pixel will be obtained, which is given in (7).

$$CF_{Hil}(x, y) = \frac{\sum_{n=1}^N |p_{Hil,n}(\Delta t_{(x,y)})|^2}{N |\sum_{n=1}^N p_{Hil,n}(\Delta t_{(x,y)})|^2} \quad (7)$$

By substituting (6) in (7), (7) is simplified as follows:

$$CF_{Hil}(x, y) = \frac{\sum_{n=1}^N |p_{Hil,n}(\Delta t_{(x,y)})|^2}{N |p_{Hil-DAS}^{(b)}(x, y)|^2} \quad (8)$$

Pixel-by-pixel multiplication of expression (6) and (8) yields the final Hilbert-coherence-based reconstructed image (DAS-H-CF) which is as follows:

$$\begin{aligned} p_{DAS-H-CF}(x, y) &= p_{Hil-DAS}^{(b)}(x, y) \times CF_{Hil}(x, y) \\ &= \frac{\sum_{n=1}^N |p_{Hil,n}(\Delta t(x, y))|^2}{N(p_{Hil-DAS}^{(b)}(x, y))}. \end{aligned} \quad (9)$$

Both the numerator and denominator of (9) are envelopes. Therefore, it is expected that the main parts of the reconstructed image will not be null, which is proven by the numerical studies.

IV. RESULTS AND DISCUSSION

A. Numerical Study

The k-Wave toolbox [21] was used for the numerical study. In numerical experiments, a 2D planar photoacoustic imaging system was considered. In the ROI, the medium was homogeneous, and the speed of sound (SOS) was 1500 m/s [22]. For the simulations to be close to reality, the properties of a commercial transducer L11-v4 were considered as numerical transducers. 128-acoustic transducers were placed on a planar array of 38 mm width. The pitch and kerf size was 0.3 and 0.03 mm, respectively. The medium grid size was 1600×1600 along both X and Y directions and the size of each grid was considered 0.025×0.025 mm in both directions. Also, a perfect match layer (PML) with 10 grids was considered in both directions. All acoustic waves (PA signals), were captured with a sampling frequency of 120 MHz. All transducers have a central frequency of 6.25 MHz and 70% bandwidth [23]. The attenuation coefficient was considered to be zero. Two types of numerical phantoms were synthesized to evaluate and compare the performance of the proposed method with the conventional methods.

Subsections B and C show the results of numerical axial point targets and different size targets, respectively. In both numerical studies, Gaussian noise with an SNR level of 25 dB was added to the captured acoustic waves.

B. Phantom1: Axial Point Targets

In axial point targets, nine-point PA targets were placed along the axial direction in two different sizes. In case 1, each absorber has a radius of 0.025 mm and the distance between absorbers is 0.25 mm. In case 2, the distance between absorbers is 0.75 mm. The radius of each absorber is 0.075 mm. The schematics of synthesized PA absorbers in both cases are shown in Fig. 2(a, b) and Fig. 3(a, b). The area within the white boxes in Fig. 2, 3(a, c, e, g) are magnified and shown in Fig. 2, 3(b, d, f, h). The reconstructed image of the DAS algorithm of cases 1 and 2 are shown in Fig. 2(c) and 2(d). Fig. 2(e, f) and Fig. 2(g, h) show the reconstructed image with the DAS algorithm multiplied by CF (DAS-CF) and Hilbert-based CF (DAS-H-CF), respectively. By comparing the results of DAS-CF and DAS-H-CF, it can be seen that in some parts of the reconstructed image of DAS-CF, some null pixels appear in the reconstructed image. But

in the reconstructed image of DAS-H-CF, a high-quality reconstructed image of the PA absorber is obtained. Quantitatively speaking, the average CR value of areas within white boxes in Fig. 2 and 3 for DAS, DAS-CF, and DAS-H-CF methods are 15, 25, and 29 dB. Moreover, the SSIM values of the same boxes are 0.73, 0.89, and 0.92, respectively.

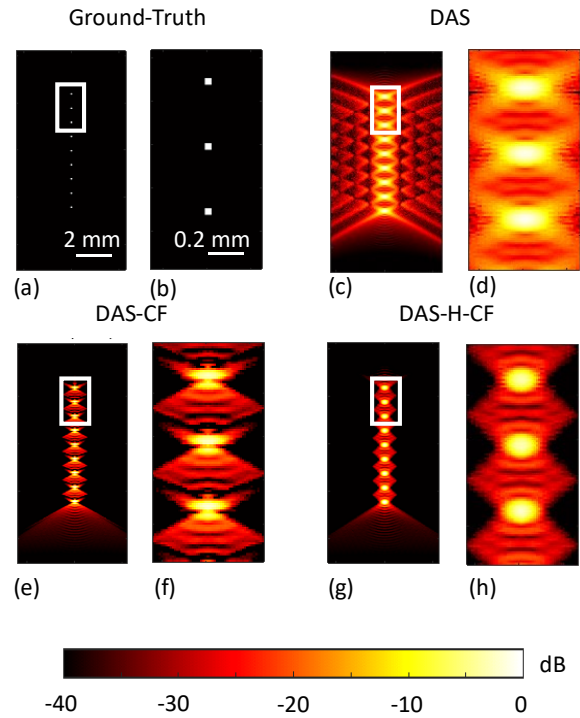


Fig. 2: The reconstructed image of case 1 of phantom 1. (a) is the schematic of the PA absorbers. (c) the reconstructed image with the DAS method. (e) and (g) are the reconstructed image with DAS-CF and DAS-H-CF methods, respectively. (b, d, f and h) are magnified areas within white boxes in (b, d, f and h), respectively.

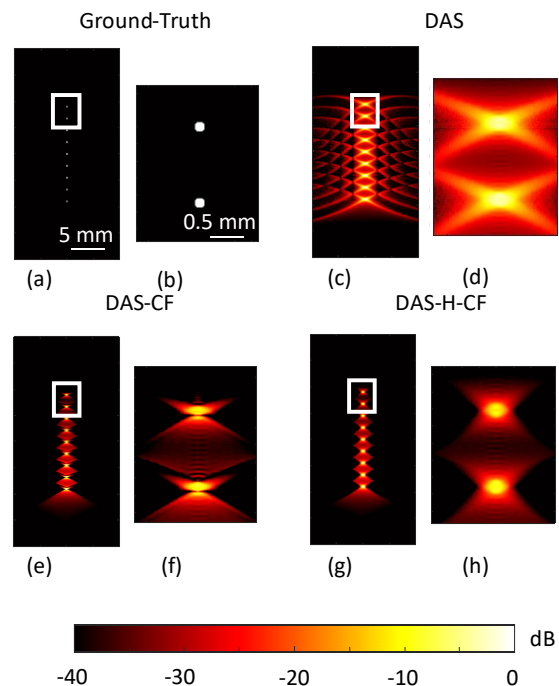


Fig. 3: The reconstructed image of case 2 of phantom 1. (a) is the schematic of the PA absorbers. (c) the reconstructed image with the DAS method. (e) and (g) are the reconstructed image with DAS-CF and DAS-H-CF methods, respectively. (b, d, f and h) are magnified areas within white boxes in (b, d, f and h), respectively.

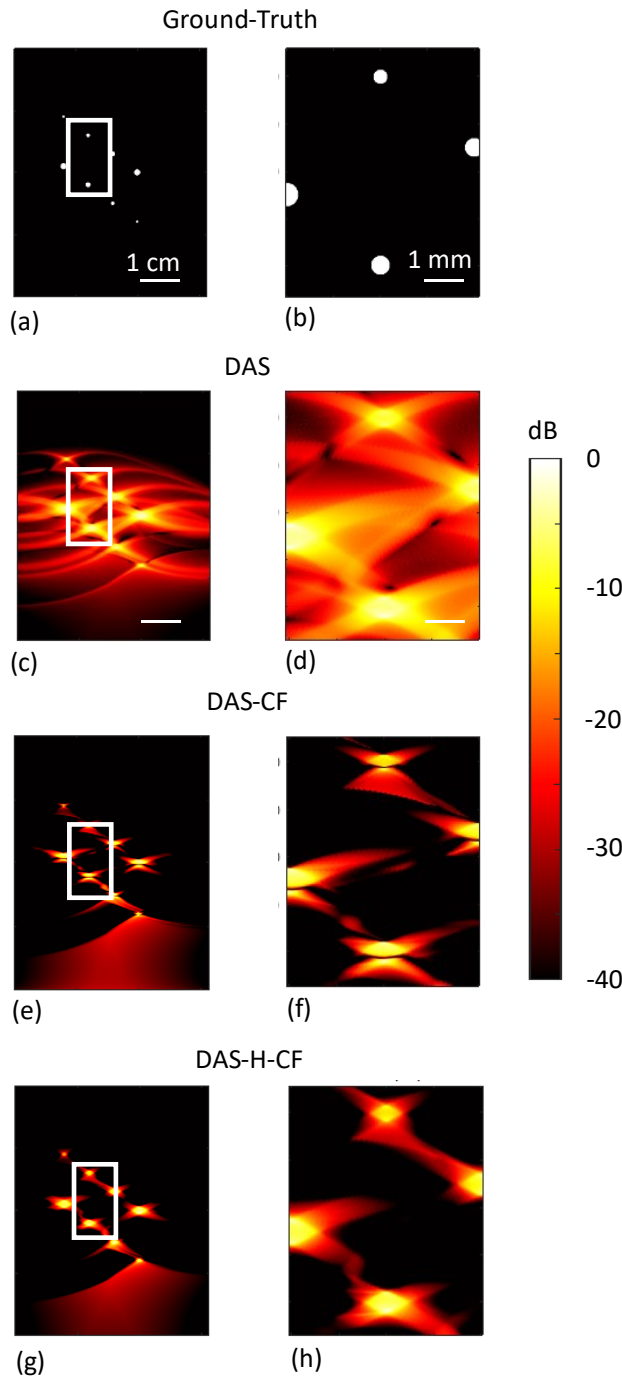


Fig. 4: The reconstructed image of case 1 of phantom 1. (a) is the schematic of the PA absorbers. (c) the reconstructed image with the DAS method. (e) and (g) are the reconstructed image with DAS-CF and DAS-H-CF methods, respectively. (b, d, f and h) are magnified areas within white boxes in (b, d, f and h), respectively.

C. Phantom2: Different Size Targets

In different size targets, eight-point PA targets were placed in the ROI. The radius of synthesized PA targets are 0.025, 0.05, 0.075, 0.1, 0.125, 0.15, 0.175 and 0.2 mm. The

schematics of synthesized PA absorbers are shown in Fig. 4(a, b). The area within the white boxes in Fig. 4(a, c, e, g) are magnified and shown in Fig. 4(b, d, f, h). The reconstructed image of the DAS algorithm is shown in Fig. 4(c, d). Also, Fig. 4(e, f) and Fig. 4(g, h) are the reconstructed images with the DAS algorithm multiplied by CF (DAS-CF) and multiplied by Hilbert-based CF (DAS-H-CF), respectively. Like phantom 1, comparing the results of DAS-CF and DAS-H-CF shows that some parts of the result of DAS-CF are missed. But in the reconstructed image of DAS-H-CF, a high-quality image of the PA absorber is reconstructed with no missing data. Quantitatively speaking, the average CR value of areas within white boxes in Fig. 2 and 3 for DAS, DAS-CF, and DAS-H-CF methods are 11, 27, and 30 dB. Moreover, the SSIM values of the same boxes are 0.71, 0.85, and 0.89, respectively.

V. CONCLUSION

To improve the resolution and contrast of the reconstructed image, coherence factor can be applied to the output of the DAS algorithm. But the results show that applying CF to the reconstructed image of the DAS algorithm results in missing data in some pixels. To tackle the missed data issue, we proposed the Hilbert-based coherence factor method (H-CF). In the proposed method, before reconstructing the image, the Hilbert transform is applied to the captured signals by the transducers. Since the newly generated signals have complex values, in DAS and CF algorithms some modifications should be considered. The results show that the SSIM and CR values in the proposed method reach ~ 0.92 and 30 dB, respectively, which are 0.19 and 19 dB more than the DAS algorithm. Since the proposed method has tackled the missed data issue and increased the resolution and contrast of the reconstructed image, it can be a suitable replacement for conventional methods.

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